

1.

$$\begin{bmatrix} 1.19 & 2.11 & -100 & 1 & 1.12 \\ 14.2 & 0.112 & 12.2 & -1 & 3.44 \\ 0 & 100 & -99.9 & 1 & 2.15 \\ 15.3 & 0.11 & -13.1 & -1 & 4.16 \end{bmatrix}$$

$$= \begin{bmatrix} 15.3 & 0.11 & -13.1 & -1 & 4.16 \\ 0 & -9.214 & 24.358 & -0.072 & -0.424 \\ 0 & 100 & -99.9 & 1 & 2.15 \\ 0 & 2.1014 & -98.982 & 1.078 & 0.9965 \end{bmatrix}$$

$$X = \begin{bmatrix} 0.1768 \\ 0.0127 \\ -0.2066 \\ -1.1833 \end{bmatrix} \#$$

2.

$$\left[\begin{array}{cccc|cccc} 4 & 1 & -10 & & 1 & 0 & 0 & 0 \\ & & & & 0 & 1 & 0 & 0 \\ 1 & 3 & -10 & & 0 & 0 & 1 & 0 \\ -1 & -1 & 6 & 2 & & & & \\ 0 & 0 & 2 & 5 & 0 & 0 & 0 & 1 \end{array} \right] \rightarrow$$

$$\left[\begin{array}{cccc|cccc} 1 & 0.25 & -0.25 & 0 & 0.25 & 0 & 0 & 0 \\ & 2.75 & -2.75 & 0 & -0.25 & 1 & 0 & 0 \\ 0 & -0.75 & 5.75 & 2 & 0.25 & 0 & 1 & 0 \\ 0 & 0 & 2 & 5 & 0 & 0 & 0 & 1 \end{array} \right] \rightarrow$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{47}{248} & -\frac{17}{62} & \frac{5}{62} & \frac{1}{124} \\ 0 & 1 & 0 & 0 & -\frac{27}{248} & \frac{67}{62} & \frac{1}{62} & -\frac{3}{124} \\ 0 & 0 & 1 & 0 & \frac{3}{124} & -\frac{1}{31} & \frac{16}{31} & -\frac{2}{31} \\ 0 & 0 & 0 & 1 & -\frac{3}{62} & \frac{1}{31} & -\frac{6}{31} & \frac{17}{62} \end{array} \right]$$

$$\text{find } A^{-1}$$

3.

$$A = LU$$

$$Ly = \begin{bmatrix} 3 & 0 & 0 & 0 \\ -1 & \frac{8}{3} & 0 & 0 \\ 0 & -1 & \frac{21}{8} & 0 \\ 0 & 0 & -1 & \frac{55}{21} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$$

$$y = \begin{bmatrix} \frac{2}{3} \\ \frac{11}{8} \\ \frac{4}{3} \\ \frac{64}{55} \end{bmatrix}, \quad Ux = \begin{bmatrix} 1 & -\frac{1}{3} & 0 & 0 \\ 0 & 1 & -\frac{3}{8} & 0 \\ 0 & 0 & 1 & -\frac{8}{21} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = y$$

$$X = \begin{bmatrix} 2.29 \\ 4.46 \\ 6.46 \\ 1.16 \end{bmatrix} \quad \#$$